AAYUSH SIGDEL

Computer Engineer | Software Developer | UI/UX Designer | System and Quality Analyst

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PROFESSIONAL SUMMARY

Registered Computer Engineer with a Bachelor of Computer Engineering from Pokhara University and hands-on experience building web applications, automated scheduling tools, and computer vision monitoring systems. Skilled in user-centered UI/UX design, system analysis, and quality assurance (functional testing, test cases, defect reporting), with a practical grounding in data science and machine learning using Python. Seeking a Software Developer, UI/UX Designer, or System and Quality Analyst role to build reliable, user-friendly solutions.

TECHNICAL SKILLS

- **Programming and Web Development:** Python, Flask, HTML, CSS, JavaScript, SQLite, web application development, system design
- **UI/UX Design:** User-centered design, interface design, wireframing, usability
- **System Analysis and Quality Assurance:** Requirements analysis, functional testing, test cases, defect reporting, validation
- **Data Science and Machine Learning:** Pandas, statistics, regression, Naive Bayes, K-Means, decision trees, XGBoost, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Principal Component Analysis (PCA), recommender systems
- **Computer Vision and Deep Learning:** PyTorch, OpenCV, activity recognition
- **Languages:** English, Nepali

PROJECTS

Classroom Activity Monitoring System

- Developed a semi-real-time system that monitors classroom activity using computer vision and deep learning techniques
- Implemented activity recognition and attendance-related computer vision functionality
- Designed role-based access for admin, teacher, and student users
- Technologies: Python, PyTorch, OpenCV, Flask, SQLite

Seat Plan Generating System

- Built a system that automatically generates examination seating arrangements, reducing manual effort and improving allocation efficiency
- Applied custom system logic and structured data handling to produce organised seat plans

E-commerce Website

- Developed a web-based shopping platform supporting product browsing and online shopping workflows
- Focused on a clean interface, clear product presentation, and smooth application functionality

EDUCATION

Bachelor of Computer Engineering 2022 - 2026

Pokhara University, School of Engineering | Dhungepatan, Nepal

CERTIFICATIONS

Registered Computer Engineer - professionally registered engineer in Nepal

Data Science and Machine Learning with Python - 6-module course covering Python and Pandas, statistics and probability, predictive models, supervised and unsupervised learning, recommender systems, KNN, and PCA